Date _____ 20__

Predicting "Rain"

	Out look	Temprature	Humidity	Windy	Rain
(1)	Sunny	Hot	high	weak	NO
(2)	Sunny	mild	high	Strong	NO
(3)	cloudy	Hot	high	weak	yes
(4)	Rainy	mild	high	weak	yes
(5)	Rainy	cool	Normal	Weak	yes
(6)	Rainy	Cool	Normal	Strong	NO
(7)	cloudy	mild	Normal	Strong	yes
(8)	Sunny	cool	Normal	Weak	NO

Step ① Total Dataset = 8, yes = 4, NO = 4

Calculating the entroppy of Rain

[label] yes no

$$-\left(\frac{4}{8}\right)\log(2)\left(\frac{4}{8}\right) + \left(\frac{4}{8}\log_2\left(\frac{4}{8}\right)\right)$$

$0.5 + 0.5 = 1$

$\Rightarrow 1$

Step ②

Information gain Of outlook:

	total	yes	no	Entroppy:
Sunny:	3	0	3	0.000
cloudy:	2	2	0	0.000
Rainy:	3	2	1	0.9631

Rainy $- \left(\frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) + \frac{1}{3}\log_2\frac{1}{3} =$ (yes)

$\Rightarrow$ [redacted] 0.917 Answer

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weighted entroppy

$\Rightarrow \frac{3}{8}(0) + \frac{2}{8} + \frac{3}{8}(0.917)$

$\Rightarrow$ 0.343

Rain – Temprature

$\Rightarrow$

$1 - 0.343 \Rightarrow 0.657$

Step ③

Temprature

	yes	NO		entroppy
Hot $\Rightarrow$	$-\left(\frac{1}{2}\log_2\frac{1}{2}\right) + \frac{1}{2}\log_2\frac{1}{2}$		$\Rightarrow$	1
Cold $\Rightarrow$	$-\left(\frac{1}{3}\log_2\frac{1}{3}\right) + \frac{2}{3}\log_2\frac{2}{3}$		$\Rightarrow$	0.917
mild $\Rightarrow$	$-\left(\frac{2}{3}\log_2\frac{2}{3}\right) + \frac{1}{3}\log_2\frac{1}{3}$		$\Rightarrow$	0.917

weighted entroppy of Temprature

$\frac{2}{8}(1) + \frac{3}{8}(0.917) + \frac{3}{8}(0.917) \Rightarrow 1.722$

$\Rightarrow$ $\Rightarrow 0.937$

$1 - 0.937$

$\Rightarrow 0.06225$ Answer

Step ③

Humidity	yes	no	entroppy
high $\Rightarrow$	$-\frac{2}{4}\log_2\frac{2}{4} + \frac{2}{4}\log_2\frac{2}{4}$	$\Rightarrow$	1
Normal $\Rightarrow$	$-\frac{2}{4}\log_2\frac{2}{4} + \frac{2}{4}\log_2\frac{2}{4}$	$\Rightarrow$	1

weighted entroppy

$\frac{4}{8}(1) + \frac{4}{8}(1) = 1$

CamScanner

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$\Rightarrow 1 - 1 = 0$ Answer

Step (4)

windy =>	Yes	no	entropy
weak =>	$-\frac{3}{5}\log_2\frac{3}{5}$	$+\frac{2}{5}\log_2\frac{2}{5}$	$\Rightarrow$ 0.970
Stron =>	$-\frac{1}{3}\log_2\frac{1}{3}$	$+\frac{2}{3}\log_2\frac{2}{3}$	$\Rightarrow$ 0.918

Weighted => $\frac{5}{8}(0.970) + \frac{3}{8}(0.918)$

$\Rightarrow 0.9505$

$\Rightarrow 1 - 0.9505 \Rightarrow 0.049$

	weighted	info gain
Outlook	0.343	0.657
Temprature	0.937	0.0625
Humidity	1	0
windy	0.9505	0.049

Outlook has the greatest information gain so It will be our root node.